

# The Eigenfunction Matching Method for a Single Floating Elastic Disc

From boundary value problem to deflection, and its extension to irregular seas

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This note walks through the forward model implemented for Phase 1 of the SHARC project: the eigenfunction matching method (EMM) of Montiel's Otago thesis for a single floating circular elastic plate with finite draught, terminating in the plate deflection  $\eta(r, \theta)$ , and its extension to a realistic irregular sea state via a JONSWAP spectrum. All notation is defined from scratch. Equation numbers in parentheses refer to Montiel's thesis unless otherwise stated.

## 1 Geometry, scaling, and the boundary value problem

A single homogeneous elastic disc of radius  $R$ , thickness  $h$ , density  $\rho$  floats on water of density  $\rho_0$  and undisturbed depth  $H$ . Archimedes gives the draught  $d = (\rho/\rho_0)h$ . Cylindrical coordinates  $\mathbf{x} = (r, \theta, z)$  are centred on the disc, with  $z = 0$  the undisturbed free surface and  $z = -H$  the bed. Two fluid regions are distinguished:

$$\Omega_0 = \{r > R, -H \leq z \leq 0\} \quad (\text{open water}), \quad \Omega = \{r \leq R, -H \leq z \leq -d\} \quad (\text{disc-covered}),$$

meeting at the vertical interface  $\Upsilon = \{r = R, -H \leq z < -d\}$ . The disc underside at rest is  $\Lambda = \{r \leq R, z = -d\}$ .

The fluid is inviscid, incompressible and irrotational, so the velocity field is  $\nabla\Phi$  for a scalar potential  $\Phi$ . The disc is modelled by Kirchhoff-Love thin-plate theory with flexural rigidity  $D = Eh^3/12(1-\nu^2)$ . Motion is assumed time-harmonic at angular frequency  $\omega$ ,  $\Phi(\mathbf{x}, t) = \text{Re}\{\phi(\mathbf{x})e^{i\omega t}\}$ , and likewise for the deflection  $\eta(r, \theta)$ .

Non-dimensionalising lengths by  $H$  and time by  $\sqrt{H/g}$  yields three parameters:

$$\alpha = \frac{H\omega^2}{g} \quad (\text{frequency}), \quad \beta = \frac{D}{\rho_0 g H^4} \quad (\text{rigidity}), \quad \gamma = \frac{d}{H} \quad (\text{draught}),$$

plus the non-dimensional radius  $R$  and Poisson's ratio  $\nu$  (fixed at 0.3). Overbars are dropped hereafter. The boundary value problem is (2.13):

$$\nabla^2 \phi = 0, \quad \mathbf{x} \in \Omega_0 \cup \Omega, \quad (\text{a})$$

$$\partial_z \phi = 0, \quad z = -1, \quad (\text{b})$$

$$\partial_z \phi = \alpha \phi, \quad \mathbf{x} \in \Lambda_0 \ (z = 0), \quad (\text{c})$$

$$(\beta \nabla_{r,\theta}^4 - \alpha \gamma + 1) \partial_z \phi = \alpha \phi, \quad \mathbf{x} \in \Lambda \ (z = -\gamma), \quad (\text{d})$$

$$[r^2 \nabla_{r,\theta}^2 - (1 - \nu)(r \partial_r + \partial_\theta^2)] \eta = 0, \quad r = R, \quad (\text{e})$$

$$[r^3 \partial_r \nabla_{r,\theta}^2 + (1 - \nu)(r \partial_r - 1) \partial_\theta^2] \eta = 0, \quad r = R, \quad (\text{f})$$

closed by a radiation condition at infinity. Condition (c) is the ordinary open-water free-surface condition: gravity is the only restoring force. Condition (d) is the same statement for the ice-covered surface, with two extra physical effects folded in: the  $\beta \nabla_{r,\theta}^4$  term is the plate's own bending stiffness (Kirchhoff-Love plate theory), and the  $-\alpha \gamma$  term is the plate's own inertia. Conditions (e)-(f) are the free-edge conditions: zero bending moment and zero shear at the rim. Surge and sway are suppressed, so additionally  $\partial_r \psi_0 = 0$  on  $\{r = R, -\gamma \leq z \leq 0\}$ . Throughout,  $\psi_0 = \phi^{\text{In}} + \phi_S$  denotes the potential in  $\Omega_0$  and  $\psi$  that in  $\Omega$ .

The key structural point: (d) is fourth order in the horizontal operator, and (b)-(d) are posed on different vertical intervals in the two regions ( $[-1, 0]$  vs  $[-1, -\gamma]$ ). Both facts drive everything that follows.

## 2 Vertical eigenfunctions and dispersion relations

Separation  $\phi = P(r, \theta)\varphi(z)$  reduces the  $z$ -dependence to a one-dimensional eigenvalue problem in each region.

**Open water  $\Omega_0$ .** Using (a)-(c),  $\psi_0 = \sum_{m \geq 0} P_m^{(0)}(r, \theta) \cos k_m(z+1)$ , with  $k_m$  the roots of

$$K \tan K = -\alpha. \quad (2.14)$$

This is a genuine Sturm-Liouville problem, so the eigenfunctions are orthogonal and complete. There is one root  $k_0$  on the positive imaginary axis (travelling mode) and infinitely many positive real roots  $k_m$ ,  $m \geq 1$ , ascending, with  $k_m \sim m\pi$  (evanescent modes). Normalising to unity at  $z = 0$ :

$$\mathcal{V}_0 = \left\{ \varphi_m^{(0)}(z) = \frac{\cos k_m(z+1)}{\cos k_m} : z \in [-1, 0], m \in \mathbb{N} \right\}. \quad (1)$$

**Disc-covered  $\Omega$ .** Using (a), (b), (d),  $\psi = \sum_{m \geq -2} P_m(r, \theta) \cos \kappa_m(z+1)$ , with  $\kappa_m$  the roots of

$$(\beta \kappa^4 + 1 - \alpha \gamma) K \tan K(1 - \gamma) = -\alpha. \quad (2.15)$$

Setting  $\beta = \gamma = 0$  collapses this exactly onto (2.14); numerically, both relations are roots of the same entire function, which is a useful implementation shortcut. Again  $\kappa_0$  is imaginary (travelling) and  $\kappa_m$ ,  $m \geq 1$ , real ascending (evanescent), with  $\kappa_m \sim m\pi/(1 - \gamma)$ . The quartic term generates two additional roots  $\kappa_{-1}, \kappa_{-2}$ : a complex-conjugate pair with positive real part. These support waves that both propagate and decay: Montiel calls them damped modes to distinguish them from purely decaying evanescent modes. They carry no far-field energy. Normalising to unity at  $z = -\gamma$  (the wetted underside):

$$\mathcal{V} = \left\{ \varphi_m(z) = \frac{\cos \kappa_m(z+1)}{\cos \kappa_m(1-\gamma)} : z \in [-1, -\gamma], m \in \mathbb{N} \right\}, \quad \varphi_{-j}(z) = \frac{\cos \kappa_{-j}(z+1)}{\cos \kappa_{-j}(1-\gamma)}, j = 1, 2. \quad (2)$$

Because  $\kappa_{-1}^2, \kappa_{-2}^2$  are not real, Sturm-Liouville theory does not apply: the full set  $\{\cos \kappa_m(z+1)\}_{m \geq -2}$  is complete but not orthogonal (Fox and Squire 1994), and moreover overcomplete.

**Bennetts reduction.** Bennetts et al. (2007) showed the set can be reduced by exactly two: each damped eigenfunction is an infinite series in the real/evanescent tower,

$$\varphi_{-j}(z) = \sum_{m=0}^{\infty} v_{m,j} \varphi_m(z), \quad j = 1, 2. \quad (2.16)$$

The coefficients follow from a residue expansion of  $1/\mathcal{D}$ , where  $\mathcal{D}$  is (2.15) written as an entire function:

$$\mathcal{D}(\xi) = (1 - \alpha \gamma + \beta \xi^4) \xi \sin \xi(1 - \gamma) + \alpha \cos \xi(1 - \gamma), \quad \xi \in \mathbb{C}, \quad (3)$$

$$\mathbf{v}_{m,j} = \frac{\kappa_m \mathcal{D}'(\kappa_{-j})(\kappa_m^2 - \kappa_{-j^*}^2)}{\kappa_{-j} \mathcal{D}'(\kappa_m)(\kappa_{-j}^2 - \kappa_{-j^*}^2)}, \quad v_{m,j} = \mathbf{v}_{m,j} \frac{\cos \kappa_m(1 - \gamma)}{\cos \kappa_{-j}(1 - \gamma)}, \quad j^* = 3 - j. \quad (4)$$

The  $\mathcal{D}'$  ratios are residues at simple zeros; the  $(\kappa_m^2 - \kappa_{-j^*}^2)$  factor removes the contribution of the other damped pole.  $\mathbf{v}_{m,j}$  is Bennetts' coefficient for the unnormalised cosines;  $v_{m,j}$  rescales it for the normalised basis (2). Assemble

$$\mathbf{V} \in \mathbb{C}^{(M+1) \times 2}, \quad [\mathbf{V}]_{m+1,j} = v_{m,j^*}, \quad (5)$$

where the  $j^* = 3 - j$  swap places column 1 against  $\kappa_{-2}$  and column 2 against  $\kappa_{-1}$ , matching the ordering used consistently throughout the rest of the pipeline.

### 3 Horizontal dependence

The polar functions are  $2\pi$ -periodic and expand in Fourier series in  $\theta$ ; substituting into the Helmholtz equation  $\nabla^2 f - K^2 f = 0$  gives the modified Bessel ODE, whose solutions are  $I_n(Kr)$  and  $K_n(Kr)$ . Boundedness at the origin selects  $I_n$  inside the disc; the radiation condition selects  $K_n$  outside. Montiel uses scaled Bessel functions

$$\hat{X}_n(Kr) = X_n(Kr)/X_n(KR) \quad (6)$$

so that every radial function is unity at  $r = R$ . Then

$$\phi_S(\mathbf{x}) \approx \sum_{m=0}^M \varphi_m^{(0)}(z) \sum_n [\mathbf{A}_n^{(0)}]_{m+1} \hat{K}_n(k_m r) e^{in\theta}, \quad \mathbf{x} \in \Omega_0, \quad (2.20)$$

$$\psi(\mathbf{x}) \approx \sum_{p=-2}^M \varphi_p(z) \sum_n A_{n,p} \hat{I}_n(\kappa_p r) e^{in\theta}, \quad \mathbf{x} \in \Omega. \quad (2.19)$$

The ambient wave is a unit-amplitude plane wave travelling along  $-x$ ,  $\phi^{\text{In}} = (1/i\alpha)e^{k_0 x} \varphi_0^{(0)}(z)$ , expanded in Bessel series as

$$[\mathbf{A}_n^{(\text{In})}]_1 = \frac{1}{i\alpha} I_n(k_0 R), \quad [\mathbf{A}_n^{(\text{In})}]_{m+1} = 0 \quad (1 \leq m \leq M). \quad (2.22)$$

Truncation:  $M+1$  vertical modes,  $2N+1$  angular modes. In matrix form, with  $\mathbf{C}_0(z)$  the row vector of  $\varphi_m^{(0)}(z)$  and  $\mathbf{E}_n^{(I)}, \mathbf{E}_n^{(0)}$  diagonal with entries  $\hat{I}_n(k_m r), \hat{K}_n(k_m r)$ ,

$$\psi_0(\mathbf{x}) \approx \mathbf{C}_0(z) \sum_n \left[ \mathbf{E}_n^{(I)}(r) \mathbf{A}_n^{(\text{In})} + \mathbf{E}_n^{(0)}(r) \mathbf{A}_n^{(0)} \right] e^{in\theta}. \quad (2.23)$$

By (6),  $\mathbf{E}_n^{(I)}(R) = \mathbf{E}_n^{(0)}(R) = \mathbf{I}_{M+1}$ .

#### 4 The free-edge map: eliminating the damped modes

The interior expansion (2.19) carries  $M+3$  amplitudes per angular order, two more than the exterior. The free-edge conditions (2.13e)-(2.13f) supply exactly two constraints per  $n$ , closing the count.

The kinematic condition on  $\Lambda$  is  $i\alpha\eta = \partial_z\psi$ , giving the deflection directly from the unreduced interior expansion:

$$\eta(r, \theta) \approx \frac{1}{i\alpha} \sum_{m=-2}^M \varphi'_m(-\gamma) \sum_n A_{n,m} \hat{I}_n(\kappa_m r) e^{in\theta}, \quad r \leq R, \quad (\text{A.2})$$

where  $\varphi'_m(-\gamma) = -\kappa_m \tan \kappa_m(1-\gamma)$ , evaluated at the plate's actual wetted edge. This is a genuinely important point: the reduction (2.16) is only valid as an  $L^2$  identity, and hence only inside the matching integrals below; the moment  $\partial_z$  is applied, the termwise-differentiated series diverges. The deflection must therefore always be built from the true  $m = -2, \dots, M$  modes with their own derivatives, never from the reduced representation.

Substituting (A.2) into (2.13e)-(2.13f) and invoking linear independence of  $\{e^{in\theta}\}$ :

$$\sum_{m=-2}^M \varphi'_m \left[ (\kappa_m R)^2 - (1-\nu)(\kappa_m R \hat{I}'_n(\kappa_m R) - n^2) \right] A_{n,m} = 0, \quad (\text{A.3a})$$

$$\sum_{m=-2}^M \varphi'_m \left[ (\kappa_m R)^3 \hat{I}'_n(\kappa_m R) - n^2(1-\nu)(\kappa_m R \hat{I}'_n(\kappa_m R) - 1) \right] A_{n,m} = 0, \quad (\text{A.3b})$$

where  $\hat{I}'_n$  denotes the derivative of  $\hat{I}_n$  with respect to its own argument. Splitting the  $m$ -sum at  $m = 0$  gives  $\bar{\mathbf{L}}_n \mathbf{A}_n + \tilde{\mathbf{L}}_n \tilde{\mathbf{A}}_n = 0$ , where  $\bar{\mathbf{L}}_n$  is  $2 \times (M+1)$  (columns  $m = 0, \dots, M$ ) and  $\tilde{\mathbf{L}}_n$  is  $2 \times 2$  (columns  $\kappa_{-2}, \kappa_{-1}$ ). Hence the edge map

$$\tilde{\mathbf{A}}_n = \mathbf{L}_n \mathbf{A}_n, \quad \mathbf{L}_n = -\tilde{\mathbf{L}}_n^{-1} \bar{\mathbf{L}}_n. \quad (\text{A.4})$$

This permits the interior potential, value only, no derivative, to be written with  $\mathbf{A}_n$  alone:

$$\psi(\mathbf{x}) \approx \mathbf{C}(z) \sum_n \bar{\mathbf{E}}_n(r) \mathbf{A}_n e^{in\theta}, \quad \bar{\mathbf{E}}_n(r) = \mathbf{E}_n(r) + \mathbf{V} \tilde{\mathbf{E}}_n(r) \mathbf{L}_n, \quad (2.24, \text{A.5})$$

with  $[\mathbf{E}_n(r)]_{m+1, m+1} = \hat{I}_n(\kappa_m r)$  and  $[\tilde{\mathbf{E}}_n(r)]_{j,j} = \hat{I}_n(\kappa_{-j} r)$ . At the interface,  $\bar{\mathbf{E}}_n(R) = \mathbf{I}_{M+1} + \mathbf{V} \mathbf{L}_n$ , an identity confirmed to hold to machine precision in the implementation.

## 5 Eigenfunction matching and the diffraction transfer matrices

**The interface unknown and its singularity.** Continuity of pressure and velocity across  $\Upsilon$  reads

$$\psi_0 = \psi, \quad \partial_r \psi_0 = \partial_r \psi = u(\theta, z) = \sum_n u_n(z) e^{in\theta} \quad \text{on } \Upsilon,$$

with  $u_n(z) \equiv 0$  for  $-\gamma < z \leq 0$  (no flow through the submerged face). Local analysis of flow around a right-angled submerged corner, confirmed numerically for an elastic ice edge by Williams and Porter (2009), gives

$$u_n(z) \sim (z + \gamma)^{-1/3} \quad \text{as } z \rightarrow -\gamma. \quad (7)$$

Expanding a singular function in smooth cosines converges very slowly, so Montiel instead builds the singularity into the basis, using weighted even Gegenbauer polynomials. With  $z^* = (z + 1)/(1 - \gamma) \in (0, 1)$  and  $\delta = 1/6$ ,

$$\mathcal{V}_g = \{C_{2p}^{(1/6)}(z^*) : p \in \mathbb{N}\}, \quad C_p^{(\delta)}(y) = \frac{2}{1 - \gamma} \frac{p!(p + \delta)\Gamma^2(\delta)}{\pi 2^{1-2\delta}\Gamma(p + 2\delta)} (1 - y^2)^{\delta-1/2} G_p^{(\delta)}(y), \quad (8)$$

$G_p^{(\delta)}$  being a Gegenbauer polynomial. The weight  $(1 - y^2)^{\delta-1/2} = (1 - y^2)^{-1/3}$  carries exactly the required singularity at  $y \rightarrow 1 \Leftrightarrow z \rightarrow -\gamma$ , and the seabed condition  $u'_n(-1) = 0$  is satisfied automatically. Writing  $u_n(z) \approx \mathbf{C}_g(z^*) \mathbf{U}^{(n)}$  with  $\mathbf{U}^{(n)}$  of length  $P + 1$  (2.26).

**Weak matching conditions.** Velocity is projected onto each region's own vertical basis over its own vertical domain; pressure is projected onto the singular basis over the shared strip only:

$$\int_{-1}^0 \mathbf{C}_0^T \partial_r \psi_0 dz = \int_{-1}^0 \mathbf{C}_0^T u dz, \quad \int_{-1}^{-\gamma} \mathbf{C}^T \partial_r \psi dz = \int_{-1}^{-\gamma} \mathbf{C}^T u dz, \quad (2.27)$$

$$\int_{-1}^{-\gamma} \mathbf{C}_g^T(z^*) \psi_0 dz = \int_{-1}^{-\gamma} \mathbf{C}_g^T(z^*) \psi dz. \quad (2.28)$$

**Inner-product matrices.** Define  $[\mathbf{D}_0]_{ij} = \int_{-1}^0 \varphi_i^{(0)} \varphi_j^{(0)} dz$  (diagonal, by orthogonality of  $\mathcal{V}_0$ ),  $[\mathbf{D}]_{ij} = \int_{-1}^{-\gamma} \varphi_i \varphi_j dz$  (full, since  $\mathcal{V}$  is not orthogonal), and the non-square  $(M + 1) \times (P + 1)$  cross-matrices  $\mathbf{D}_g^{(0)}$  (between  $\mathcal{V}_0$  and  $\mathcal{V}_g$ ) and  $\mathbf{D}_g$  (between  $\mathcal{V}$  and  $\mathcal{V}_g$ ). Direct integration of (2) gives

$$[\mathbf{D}]_{ii} = \frac{1}{2} \left[ \frac{\tan \kappa_i(1-\gamma)}{\kappa_i} + (1 - \gamma) \sec^2 \kappa_i(1 - \gamma) \right], \quad [\mathbf{D}]_{ij} = \frac{\kappa_i \tan \kappa_i(1 - \gamma) - \kappa_j \tan \kappa_j(1 - \gamma)}{\kappa_i^2 - \kappa_j^2},$$

with  $\mathbf{D}_0$  recovered by setting  $\gamma = 0$ . The Gegenbauer cross-matrices evaluate in closed form via the degenerate Gegenbauer addition theorem:

$$[\mathbf{D}_g]_{i+1,j+1} = \Gamma(\delta) \left( \frac{2}{\kappa_i(1-\gamma)} \right)^\delta (2j + \delta) (-1)^j \frac{J_{2j+\delta}(\kappa_i(1-\gamma))}{\cos \kappa_i(1-\gamma)}, \quad (9)$$

and similarly for  $\mathbf{D}_g^{(0)}$  with  $k_i, (1 - \gamma) \rightarrow 1, \cos k_i$ .

**The linear system.** Substituting (2.23) and (2.24) into (2.27)-(2.28):

$$\mathbf{D}_0 [\mathbf{E}_n^{(I)'}(R) \mathbf{A}_n^{(\text{In})} + \mathbf{E}_n^{(0)'}(R) \mathbf{A}_n^{(0)}] = \mathbf{D}_g^{(0)} \mathbf{U}^{(n)}, \quad (B.1a)$$

$$\mathbf{D} \bar{\mathbf{E}}_n'(R) \mathbf{A}_n = \mathbf{D}_g \mathbf{U}^{(n)}, \quad (B.1b)$$

$$\mathbf{D}_g^{(0)T} [\mathbf{A}_n^{(\text{In})} + \mathbf{A}_n^{(0)}] = \mathbf{D}_g^T \bar{\mathbf{E}}_n(R) \mathbf{A}_n. \quad (B.1c)$$

This is  $2M + P + 3$  equations in the same number of unknowns ( $\mathbf{A}_n^{(0)}$ ,  $\mathbf{A}_n$ ,  $\mathbf{U}^{(n)}$ ). Note that  $\bar{\mathbf{E}}_n$ , and therefore  $\mathbf{L}_n$ , appears inside the matching system: the free-edge map is a prerequisite for the matching step, not a parallel calculation.

Eliminating via (B.1a)-(B.1b) with

$$\mathbf{M}_n^{(\text{In})} = -[\mathbf{E}_n^{(0)'}(R)]^{-1} \mathbf{E}_n^{(I)'}(R), \quad \mathbf{M}_n^{(0)} = [\mathbf{D}_0 \mathbf{E}_n^{(0)'}(R)]^{-1} \mathbf{D}_g^{(0)}, \quad \mathbf{M}_n = [\mathbf{D} \bar{\mathbf{E}}_n'(R)]^{-1} \mathbf{D}_g,$$

so  $\mathbf{A}_n^{(0)} = \mathbf{M}_n^{(\text{In})} \mathbf{A}_n^{(\text{In})} + \mathbf{M}_n^{(0)} \mathbf{U}^{(n)}$  and  $\mathbf{A}_n = \mathbf{M}_n \mathbf{U}^{(n)}$ ; substituting into (B.1c),

$$\mathbf{U}^{(n)} = \mathbf{M}_n^{(U)} \mathbf{A}_n^{(\text{In})}, \quad \mathbf{M}_n^{(U)} = \left[ \mathbf{D}_g^T \bar{\mathbf{E}}_n(R) \mathbf{M}_n - \mathbf{D}_g^{(0)T} \mathbf{M}_n^{(0)} \right]^{-1} \mathbf{D}_g^{(0)T} [\mathbf{I}_{M+1} + \mathbf{M}_n^{(\text{In})}]. \quad (10)$$

Finally,

$$\mathbf{A}_n = \mathbf{S}_n \mathbf{A}_n^{(\text{In})}, \quad \mathbf{S}_n = \mathbf{M}_n \mathbf{M}_n^{(U)}; \quad \mathbf{A}_n^{(0)} = \mathbf{S}_n^{(0)} \mathbf{A}_n^{(\text{In})}, \quad \mathbf{S}_n^{(0)} = \mathbf{M}_n^{(\text{In})} + \mathbf{M}_n^{(0)} \mathbf{M}_n^{(U)}. \quad (2.29)$$

$\mathbf{S}_n$  and  $\mathbf{S}_n^{(0)}$  are the interior and exterior diffraction transfer matrices. Crucially they depend only on  $(\alpha, \beta, \gamma, R, \nu)$ , not on the incident wave, so they are computed once per frequency and reused across every forcing realisation. This last property is the entire basis for the JONSWAP extension described in Section 8.

## 6 Deflection

$$\eta(r, \theta) \approx \frac{1}{i\alpha} \sum_{m=-2}^M \left[ -\kappa_m \tan \kappa_m (1 - \gamma) \right] \sum_{n=-N}^N A_{n,m} \hat{I}_n(\kappa_m r) e^{in\theta}, \quad r \leq R, \quad (11)$$

where  $A_{n,m}$  for  $m = 0, \dots, M$  are the entries of  $\mathbf{A}_n$  from (2.29), and  $A_{n,-2}, A_{n,-1}$  are the entries of  $\tilde{\mathbf{A}}_n = \mathbf{L}_n \mathbf{A}_n$  from (A.4). This is the only place in the pipeline where  $\bar{\mathbf{E}}_n$  must not be used, since it is only valid for potential values, not derivatives.

## 7 Pipeline summary

Per frequency  $\alpha$  and angular order  $n$ , with plate parameters  $(\beta, \gamma, R, \nu)$  fixed:

- 0. Roots and bases.** Solve (2.14) for  $k_m$ ,  $m = 0..M$ ; solve (2.15) for  $\kappa_m$ ,  $m = -2..M$ . Build  $\varphi_m^{(0)}$ ,  $\varphi_m$  and the edge derivatives  $\varphi'_m(-\gamma)$ .
- 1. Bennetts coefficients.** Evaluate  $\mathcal{D}, \mathcal{D}'$ ; form  $\mathbf{v}_{m,j}$  then  $v_{m,j}$  (4); assemble  $\mathbf{V}$  with the  $j^*$  swap.
- 2. Edge map.** Build  $\bar{\mathbf{L}}_n, \tilde{\mathbf{L}}_n$  from (A.3); invert the  $2 \times 2$  block.
- 3. Corrected radial matrix.**  $\bar{\mathbf{E}}_n(r) = \mathbf{E}_n(r) + \mathbf{V} \tilde{\mathbf{E}}_n(r) \mathbf{L}_n$ .
- 4. Matching.** Build  $\mathbf{D}_0, \mathbf{D}, \mathbf{D}_g^{(0)}, \mathbf{D}_g$ ; form  $\mathbf{M}_n^{(\text{In})}, \mathbf{M}_n^{(0)}, \mathbf{M}_n, \mathbf{M}_n^{(U)}$ ; obtain  $\mathbf{S}_n, \mathbf{S}_n^{(0)}$ .
- 5. Forcing in.** Build  $\mathbf{A}_n^{(\text{In})}$  from (2.22); compute  $\mathbf{A}_n = \mathbf{S}_n \mathbf{A}_n^{(\text{In})}$ .
- 6. Damped amplitudes.**  $\tilde{\mathbf{A}}_n = \mathbf{L}_n \mathbf{A}_n$ .
- 7. Deflection.** Sum  $m = -2, \dots, M$  with the true  $\varphi'_m(-\gamma)$ ; sum over  $n$ .

Steps 0-4 are forcing-independent and cache per  $(\alpha, n)$ ; steps 5-7 are cheap and run per realisation.

**Truncation guidance.**  $P < M$  is required or  $\mathbf{D}_g^{(0)}, \mathbf{D}_g$  become ill-conditioned.  $M \approx 20$ -30 gives two significant digits at moderate  $\beta$ ; softer plates (smaller  $\beta$ ) require larger  $M$ , confirmed empirically by doubling checks before use.  $N$  is governed by the product  $\kappa R$ , not by plate properties:  $N \approx 10$  at  $\alpha = 10$ ,  $N \approx 20$  at  $\alpha = 20$  for  $R = 1$ ; this too was confirmed by direct convergence checks rather than assumed.

**Validation performed.** The truncated  $\sum_m v_{m,j} \varphi_m(z)$  was checked against the direct  $\varphi_{-j}(z)$  at both the plate edge and interior points, confirming clean convergence as  $M$  grows. The identity  $\bar{\mathbf{E}}_n(R) = \mathbf{I} + \mathbf{V} \mathbf{L}_n$  was confirmed to machine precision. Most importantly, the recovered  $(\mathbf{A}_n, \tilde{\mathbf{A}}_n)$  were substituted back into (A.3a)-(A.3b): both residuals returned at machine precision ( $\sim 10^{-16}$ ), confirming the entire chain from roots through scattering matrices is jointly self-consistent. External validation against an independent model (the  $\beta \rightarrow \infty$  rigid limit, the  $\gamma \rightarrow 0$  shallow-draught collapse, and a genuine energy-conservation check) remains outstanding.

## 8 Extension to irregular seas: JONSWAP forcing

The model above computes the response to a single monochromatic incident wave. Real sea states are irregular, commonly described by a JONSWAP spectrum,  $S(\omega)$ , giving the energy density as a function of angular frequency. The extension from one frequency to a full spectrum relies on a single structural fact already noted above: the boundary value problem is linear throughout, so the diffraction transfer matrices  $\mathbf{S}_n, \mathbf{S}_n^{(0)}$ , and hence the deflection, depend linearly on the incident wave amplitude. The response to a sum of frequency components is therefore exactly the sum of the responses to each component individually: no new physics is required, only a systematic way of combining many single-frequency solves.

**Discretising the spectrum.** A continuous spectrum is converted into a finite set of components using the standard random-phase and amplitude method. Choosing  $N_f$  frequency bins of width  $\Delta\omega$  centred at  $\omega_i$ , each component is assigned

$$a_i = \sqrt{2S(\omega_i)\Delta\omega}, \quad \epsilon_i \sim \text{Uniform}[0, 2\pi), \quad (12)$$

independently for each  $i$ . This follows from requiring the discrete sum to carry the same variance the continuous spectrum assigns to each frequency band: a cosine component of amplitude  $a_i$  with a uniformly random phase has mean-square value  $a_i^2/2$ , and setting this equal to  $S(\omega_i)\Delta\omega$  gives the formula above. The random phases are what make the resulting time series look like a genuine irregular sea rather than a periodic beat pattern, and are exactly why real sea waves group into sets of larger and smaller crests: components at different frequencies drift in and out of phase alignment as time progresses.

**Per-frequency reuse.** For each discretised component, the corresponding non-dimensional frequency is  $\alpha_i = H\omega_i^2/g$ , and Steps 0-6 of the pipeline above are run exactly as for a single monochromatic wave, entirely unmodified, to obtain that component's own complex deflection  $\eta_i(r, \theta)$ . Nothing in the core model changes; each frequency component is simply an independent instance of the existing single-frequency solve.

**Recombination.** The physical, time-varying deflection is then

$$\zeta(r, \theta, t) = \sum_{i=1}^{N_f} a_i \text{Re}\{\eta_i(r, \theta) e^{i(\omega_i t + \epsilon_i)}\}, \quad (13)$$

where  $t$  is real physical time. Since  $\eta_i$  is the response to a unit non-dimensional incident amplitude, and  $a_i$  is a genuine physical amplitude in metres, this expression already carries the correct physical units throughout: the redimensionalisation by  $H$  that would otherwise be needed cancels exactly against the non-dimensionalisation of  $a_i$ , a direct consequence of the model's linearity noted above.

**Fixed truncation across the spectrum.**  $M, P, N$  are held fixed across every frequency component, rather than adapted per component, for simplicity. This choice was validated by a doubling check at the highest frequency present in a given spectrum, since higher frequency generally demands larger truncation; every component at a lower frequency is then automatically covered by the same, or a more generous, truncation. This was demonstrated at wave-basin scale, matching an existing wave-tank experiment reported in Montiel's thesis, before attempting a realistic ocean-scale scenario, since realistic Antarctic water depths and wave periods push the non-dimensional frequency  $\alpha$  (which is structurally the same quantity as the deep-water parameter  $\kappa H$ ) well beyond anything validated so far, and would require the truncation question to be revisited at that scale.

**Validation performed.** The discretisation was confirmed to recover the target significant wave height,  $H_s = 4\sqrt{m_0}$ , both as  $N_f$  grows and by comparison against direct numerical integration of the continuous spectrum. The recombination step was checked three ways: against a single, near-monochromatic spectral component collapsing exactly onto the core single-frequency model; against a hand-computed multi-component sum; and by confirming that the deflection at the exact centre of the plate remains independent of angle at every instant in time, a structural consequence of the underlying Bessel functions that must hold for every frequency component individually, and therefore for their sum.

**Looking ahead.** The forcing-independent caching noted at the end of Section 5 is what makes this extension computationally practical: the diffraction transfer matrices are the expensive part of the calculation, and are computed once per frequency component regardless of how many spatial points or time steps are subsequently evaluated. This same synthetic, fully-known irregular-wave deflection time series is the intended input for the planned twin-test validation of the system identification stage: since every mode and amplitude that produced a given synthetic signal is known exactly, it provides a known-pole test case against which the identification method can be checked before it is applied to real buoy data.